

Zion Edge — Baseline Report

Rigorous benchmark · RFC conformance · cache-correctness · zion 0.4.2 · 2026-06-25T21:23:37Z · mode=full

Generated by benchmarks/baseline/run-baseline.sh. Throughput is the **median of 3 trials** with 95% CI; latency is p50/p99/p99.9 (ms). Every figure is parsed from a raw output embedded in the Appendix.

1. Summary

145/146 HTTP/2 RFC	PASS CACHE CORRECTNESS	PASS TLS VULNS	55,873 CACHE REQ/S (MEDIAN)	n/a% HIT RATIO (90/10)
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2. Environment

Zion / git	zion 0.4.2 · bdb204d
Host	Intel(R) Core(TM) i7-6700 CPU @ 3.40GHz · 3 cores · 2 GB · Linux 6.17.4-2-pve/x86_64
CPU governor / isolation	performance · governor=performance
Negotiated ALPN	zion: HTTP/2 · nginx: HTTP/2
Tools	oha 1.14.0 · h2load nghttp2/1.59.0 · wrk debian/4.1.0-4build2 [epoll] Copyright (C) 2012 Will Glozer · Version: 2.6.0 (70ac2294010887f48b18e2d64f5cccd48421fad1) · testssl.sh version 3.3dev from https://testssl.sh/dev/

3. Functional — cache correctness (validates v0.4.2)

Check	Result
Age header present on hit	PASS
Age monotonic (0→2s over 2s)	PASS
Origin TTL honoured (max-age=5 → 'public, max-age=5')	PASS
Stale-born passthrough (upstream Age=99999)	PASS
Hit ratio under 90/10 hot/cold load	n/a% (0 hits / 0 misses)

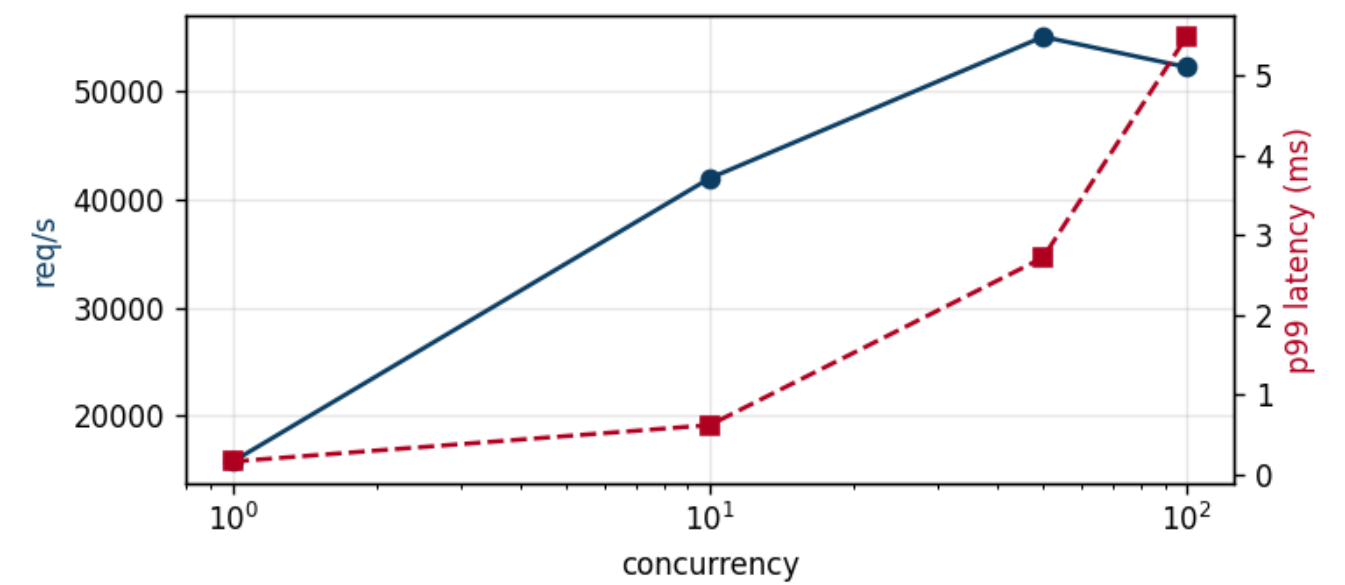
4. Throughput & latency

oha over ALPN-negotiated HTTP/2; median of 3 trials, c=50, 15s.

Target	req/s (median ±CI)	p50 ms	p99 ms	p99.9 ms	CPU	RSS MB	req/s-core	n
zion cache-hit	55,873 ±2,647	0.76	2.80	5.41	134%	58	18,624	3
zion proxy passthrough	27,666 ±169	1.68	4.44	6.48	123%	58	9,222	3
nginx cache-hit (ref)	39,531 ±603	1.06	4.13	8.11	0%	8	13,177	3

Protocol-pinned cross-check: h2load (explicit H2) 56,206 req/s (200000 ok) · wrk (explicit H1) 59,550 req/s · wrk2 CO-corrected p99 160.64ms.

5. Concurrency sweep (zion cache-hit)



concurrency	req/s	p99 ms
1	15,776	0.16
10	41,963	0.62
50	55,044	2.72
100	52,251	5.49

6. Payload matrix (zion cache-hit)

body size	req/s (median)	95% CI	p99 ms
1 KB	65,799	±5,530	2.42
64 KB	16,316	±3,209	11.05
1 MB	1,880	±97	70.78

7. Regression

no history baseline yet (first run on this host)

8. RFC conformance

HTTP/2 (RFC 9113/7540) — h2spec

total	passed	skipped	failed
146	145	0	1

- Sends invalid connection preface

TLS — testssl.sh

ALPN	h2, http/1.1
Forward Secrecy	yes

Lab-cert artifacts (self-signed, not a zion weakness): cert_chain_of_trust: failed (self signed).; cert_revocation: Neither CRL nor OCSP URI provided; overall_grade: T

Heartbleed (CVE-2014-0160) not vulnerable (OK), no heartbeat extension

CCS (CVE-2014-0224) not vulnerable (OK)

Ticketbleed (CVE-2016-9244), experimental not vulnerable (OK)

Opossum (CVE-2025-49812) not vulnerable (OK)

Secure Client-Initiated Renegotiation not vulnerable (OK) -- mitigated (1 successful renegotiation within 10 in 33s(timeout))

CRIME, TLS (CVE-2012-4929) not vulnerable (OK)

POODLE, SSL (CVE-2014-3566) not vulnerable (OK), no SSLv3 support

SWEET32 (CVE-2016-2183, CVE-2016-6329) not vulnerable (OK)

FREAK (CVE-2015-0204) not vulnerable (OK)

DROWN (CVE-2016-0800, CVE-2016-0703) not vulnerable on this host and port (OK)

LOGJAM (CVE-2015-4000), experimental not vulnerable (OK): no DH EXPORT ciphers, no DH key detected with <= TLS 1.2

BEAST (CVE-2011-3389) not vulnerable (OK), no SSL3 or TLS1

LUCKY13 (CVE-2013-0169), experimental not vulnerable (OK)

Winshock (CVE-2014-6321), experimental not vulnerable (OK)

Appendix — raw output

results/cache-correctness.txt

```
age_present=yes
age_t0=0
age_t2=2
age_monotonic=yes
shortttl_cache_control=public, max-age=5
origin_ttl_honored=yes
staleborn_age=99999
staleborn_passthrough=yes
hits_delta=0
misses_delta=0
hit_ratio=n/a
```

results/hl-zion-cache.dat

```
55010.7136 0.7794 2.7965 5.4105 30.2853 134.3 58.4
55872.6805 0.7621 2.9825 7.1824 26.0021 132.8 58.4
60348.0580 0.7280 2.4140 3.9697 9.7212 139.3 58.4
```

results/hl-zion-proxy.dat

```
27751.1898 1.6785 4.4308 6.4813 65.7725 123.1 58.4
27400.9306 1.7006 4.6011 7.1499 60.1863 122.3 58.4
27665.5933 1.6849 4.4447 6.4192 60.0231 122.9 58.4
```

results/hl-nginx-cache.dat

```
39733.2277 1.0563 3.9900 7.0352 18.0702 0.0 8.2
39530.5758 1.0006 4.3186 9.2285 61.5139 0.0 8.2
38515.0199 1.0593 4.1271 8.1139 87.0426 0.0 8.2
```

results/sweep.dat

```
1 15776.0618 0.1636
10 41962.7671 0.6198
50 55044.3246 2.7196
100 52250.6881 5.4907
```

results/h2load.txt

```
starting benchmark...
spawning thread #0: 50 total client(s). 200000 total requests
TLS Protocol: TLSv1.3
Cipher: TLS_AES_256_GCM_SHA384
Server Temp Key: X25519 253 bits
Application protocol: h2
progress: 10% done
progress: 20% done
progress: 30% done
progress: 40% done
progress: 50% done
progress: 60% done
progress: 70% done
progress: 80% done
progress: 90% done
progress: 100% done

finished in 3.56s, 56205.78 req/s, 222.79MB/s
requests: 200000 total, 200000 started, 200000 done, 200000 succeeded, 0 failed, 0 errored, 0 timeout
status codes: 200000 2xx, 0 3xx, 0 4xx, 0 5xx
traffic: 792.77MB (831278590) total, 8.08MB (8475840) headers (space savings 89.79%), 781.25MB (819200000) data

time for request:      min      max      mean      sd      +/- sd
time for request:    62us    124.70ms    9.74ms    6.21ms    95.02%
time for connect:    15.84ms    45.19ms    31.72ms    8.54ms    62.00%
time to 1st byte:    51.70ms    65.19ms    59.34ms    4.06ms    60.00%
req/s                :    1124.73    1156.35    1132.51     7.50    68.00%
```

results/wrk.txt

```
Running 15s test @ https://127.0.0.1:4432/_next/static/chunk.js
4 threads and 50 connections
Thread Stats Avg Stdev Max +/- Stdev
Latency 843.74us 1.00ms 34.66ms 94.61%
Req/Sec 14.96k 2.20k 19.24k 81.17%
894133 requests in 15.01s, 3.80GB read
Requests/sec: 59549.66
Transfer/sec: 259.19MB
```

results/wrk2.txt

```
Running 15s test @ https://127.0.0.1:4432/_next/static/chunk.js
4 threads and 50 connections
Thread calibration: mean lat.: 2.878ms, rate sampling interval: 10ms
Thread calibration: mean lat.: 4.520ms, rate sampling interval: 26ms
Thread calibration: mean lat.: 3.725ms, rate sampling interval: 17ms
Thread calibration: mean lat.: 3.759ms, rate sampling interval: 17ms
Thread Stats Avg Stdev Max +/- Stdev
Latency 16.58ms 38.67ms 172.80ms 88.40%
Req/Sec 12.98k 2.93k 22.50k 83.85%
Latency Distribution (HdrHistogram - Recorded Latency)
50.000% 1.61ms
75.000% 3.19ms
90.000% 74.69ms
99.000% 160.64ms
99.900% 168.45ms
99.990% 171.52ms
99.999% 172.67ms
100.000% 172.93ms

Detailed Percentile spectrum:
Value Percentile TotalCount 1/(1-Percentile)
0.044 0.000000 2 1.00
0.636 0.100000 24752 1.11
0.899 0.200000 49451 1.25
1.129 0.300000 74155 1.43
1.361 0.400000 98832 1.67
```

1.609	0.500000	123457	2.00
1.755	0.550000	135850	2.22
1.930	0.600000	148163	2.50
2.153	0.650000	160565	2.86
2.477	0.700000	172827	3.33
3.189	0.750000	185172	4.00
4.171	0.775000	191364	4.44
6.215	0.800000	197522	5.00
9.815	0.825000	203696	5.71
18.191	0.850000	209862	6.67
44.575	0.875000	216038	8.00
59.615	0.887500	219123	8.89
74.687	0.900000	222206	10.00
88.703	0.912500	225295	11.43
104.383	0.925000	228381	13.33
118.399	0.937500	231473	16.00
124.799	0.943750	233020	17.78
129.983	0.950000	234563	20.00
133.887	0.956250	236116	22.86
137.727	0.962500	237651	26.67
141.311	0.968750	239200	32.00
143.231	0.971875	239959	35.56
145.663	0.975000	240742	40.00
148.095	0.978125	241496	45.71
151.039	0.981250	242276	53.33
154.623	0.984375	243043	64.00
156.543	0.985938	243453	71.11
158.079	0.987500	243826	80.00
159.487	0.989062	244202	91.43
161.151	0.990625	244580	106.67
162.431	0.992188	244981	128.00
162.943	0.992969	245173	142.22
163.583	0.993750	245350	160.00
164.351	0.994531	245581	182.86
164.863	0.995313	245744	213.33
165.503	0.996094	245938	256.00
165.887	0.996484	246076	284.44
166.143	0.996875	246155	320.00
166.399	0.997266	246219	365.71
166.783	0.997656	246327	426.67
167.167	0.998047	246414	512.00
167.423	0.998242	246472	568.89
167.679	0.998437	246514	640.00
167.935	0.998633	246566	731.43
168.191	0.998828	246610	853.33
168.447	0.999023	246654	1024.00
168.575	0.999121	246678	1137.78
168.831	0.999219	246710	1280.00
168.959	0.999316	246725	1462.86
169.215	0.999414	246749	1706.67
169.599	0.999512	246780	2048.00
169.727	0.999561	246794	2275.56
169.855	0.999609	246804	2560.00
169.983	0.999658	246811	2925.71
170.239	0.999707	246824	3413.33
170.495	0.999756	246836	4096.00
170.623	0.999780	246844	4551.11
170.751	0.999805	246848	5120.00
170.879	0.999829	246851	5851.43
171.135	0.999854	246860	6826.67
171.263	0.999878	246865	8192.00
171.391	0.999890	246867	9102.22
171.519	0.999902	246871	10240.00
171.647	0.999915	246878	11702.86
171.647	0.999927	246878	13653.33
171.647	0.999939	246878	16384.00
171.903	0.999945	246881	18204.44
171.903	0.999951	246881	20480.00
172.031	0.999957	246884	23405.71
172.031	0.999963	246884	27306.67
172.159	0.999969	246886	32768.00
172.287	0.999973	246888	36408.89
172.287	0.999976	246888	40960.00
172.287	0.999979	246888	46811.43
172.415	0.999982	246890	54613.33
172.415	0.999985	246890	65536.00
172.415	0.999986	246890	72817.78
172.415	0.999988	246890	81920.00
172.671	0.999989	246892	93622.86
172.671	0.999991	246892	109226.67
172.671	0.999992	246892	131072.00
172.671	0.999993	246892	145635.56

172.671	0.999994	246892	163840.00
172.671	0.999995	246892	187245.71
172.671	0.999995	246892	218453.33
172.927	0.999996	246893	262144.00
172.927	1.000000	246893	inf
#[Mean	=	16.585, StdDeviation	= 38.666]
#[Max	=	172.800, Total count	= 246893]
#[Buckets	=	27, SubBuckets	= 2048]

748468 requests in 15.00s, 3.18GB read
 Requests/sec: 49905.64
 Transfer/sec: 217.22MB

results/h2spec.txt

Generic tests for HTTP/2 server

1. Starting HTTP/2

- 1: Sends a client connection preface
- ✓ 1: Sends a client connection preface

2. Streams and Multiplexing

- 1: Sends a PRIORITY frame on idle stream
- ✓ 1: Sends a PRIORITY frame on idle stream
- 2: Sends a WINDOW_UPDATE frame on half-closed (remote) stream
- ✓ 2: Sends a WINDOW_UPDATE frame on half-closed (remote) stream
- 3: Sends a PRIORITY frame on half-closed (remote) stream
- ✓ 3: Sends a PRIORITY frame on half-closed (remote) stream
- 4: Sends a RST_STREAM frame on half-closed (remote) stream
- ✓ 4: Sends a RST_STREAM frame on half-closed (remote) stream
- 5: Sends a PRIORITY frame on closed stream
- ✓ 5: Sends a PRIORITY frame on closed stream

3. Frame Definitions

3.1. DATA

- 1: Sends a DATA frame
- ✓ 1: Sends a DATA frame
- 2: Sends multiple DATA frames
- ✓ 2: Sends multiple DATA frames
- 3: Sends a DATA frame with padding
- ✓ 3: Sends a DATA frame with padding

3.2. HEADERS

- 1: Sends a HEADERS frame
- ✓ 1: Sends a HEADERS frame
- 2: Sends a HEADERS frame with padding
- ✓ 2: Sends a HEADERS frame with padding
- 3: Sends a HEADERS frame with priority
- ✓ 3: Sends a HEADERS frame with priority

3.3. PRIORITY

- 1: Sends a PRIORITY frame with priority 1
- ✓ 1: Sends a PRIORITY frame with priority 1
- 2: Sends a PRIORITY frame with priority 256
- ✓ 2: Sends a PRIORITY frame with priority 256
- 3: Sends a PRIORITY frame with stream dependency
- ✓ 3: Sends a PRIORITY frame with stream dependency
- 4: Sends a PRIORITY frame with exclusive
- ✓ 4: Sends a PRIORITY frame with exclusive
- 5: Sends a PRIORITY frame for an idle stream, then send a HEADER frame for a lower stream ID
- ✓ 5: Sends a PRIORITY frame for an idle stream, then send a HEADER frame for a lower stream ID

3.4. RST_STREAM

- 1: Sends a RST_STREAM frame
- ✓ 1: Sends a RST_STREAM frame

3.5. SETTINGS

- 1: Sends a SETTINGS frame
- ✓ 1: Sends a SETTINGS frame

3.7. PING

- 1: Sends a PING frame
- ✓ 1: Sends a PING frame

3.8. GOAWAY

- 1: Sends a GOAWAY frame
- ✓ 1: Sends a GOAWAY frame

3.9. WINDOW_UPDATE

- 1: Sends a WINDOW_UPDATE frame with stream ID 0
- ✓ 1: Sends a WINDOW_UPDATE frame with stream ID 0

- 2: Sends a WINDOW_UPDATE frame with stream ID 1
- ✓ 2: Sends a WINDOW_UPDATE frame with stream ID 1

3.10. CONTINUATION

- 1: Sends a CONTINUATION frame
- ✓ 1: Sends a CONTINUATION frame
- 2: Sends multiple CONTINUATION frames
- ✓ 2: Sends multiple CONTINUATION frames

4. HTTP Message Exchanges

- 1: Sends a GET request
- ✓ 1: Sends a GET request
- 2: Sends a HEAD request
- ✓ 2: Sends a HEAD request
- 3: Sends a POST request
- ✓ 3: Sends a POST request
- 4: Sends a POST request with trailers
- ✓ 4: Sends a POST request with trailers

5. HPACK

- 1: Sends a indexed header field representation
- ✓ 1: Sends a indexed header field representation
- 2: Sends a literal header field with incremental indexing - indexed name
- ✓ 2: Sends a literal header field with incremental indexing - indexed name
- 3: Sends a literal header field with incremental indexing - indexed name (with Huffman coding)
- ✓ 3: Sends a literal header field with incremental indexing - indexed name (with Huffman coding)
- 4: Sends a literal header field with incremental indexing - new name
- ✓ 4: Sends a literal header field with incremental indexing - new name
- 5: Sends a literal header field with incremental indexing - new name (with Huffman coding)
- ✓ 5: Sends a literal header field with incremental indexing - new name (with Huffman coding)
- 6: Sends a literal header field without indexing - indexed name
- ✓ 6: Sends a literal header field without indexing - indexed name
- 7: Sends a literal header field without indexing - indexed name (with Huffman coding)
- ✓ 7: Sends a literal header field without indexing - indexed name (with Huffman coding)
- 8: Sends a literal header field without indexing - new name
- ✓ 8: Sends a literal header field without indexing - new name
- 9: Sends a literal header field without indexing - new name (huffman encoded)
- ✓ 9: Sends a literal header field without indexing - new name (huffman encoded)
- 10: Sends a literal header field never indexed - indexed name
- ✓ 10: Sends a literal header field never indexed - indexed name
- 11: Sends a literal header field never indexed - indexed name (huffman encoded)
- ✓ 11: Sends a literal header field never indexed - indexed name (huffman encoded)
- 12: Sends a literal header field never indexed - new name
- ✓ 12: Sends a literal header field never indexed - new name
- 13: Sends a literal header field never indexed - new name (huffman encoded)
- ✓ 13: Sends a literal header field never indexed - new name (huffman encoded)
- 14: Sends a dynamic table size update
- ✓ 14: Sends a dynamic table size update
- 15: Sends multiple dynamic table size update
- ✓ 15: Sends multiple dynamic table size update

Hypertext Transfer Protocol Version 2 (HTTP/2)

3. Starting HTTP/2

3.5. HTTP/2 Connection Preface

- 1: Sends client connection preface
- ✓ 1: Sends client connection preface
- 2: Sends invalid connection preface
- using source address 127.0.0.1:54762
- × 2: Sends invalid connection preface
- The endpoint MUST terminate the TCP connection.
- Expected: GOAWAY Frame (Error Code: PROTOCOL_ERROR)
- Connection closed
- Actual: Error: unexpected EOF

4. HTTP Frames

4.1. Frame Format

- 1: Sends a frame with unknown type
- ✓ 1: Sends a frame with unknown type
- 2: Sends a frame with undefined flag
- ✓ 2: Sends a frame with undefined flag
- 3: Sends a frame with reserved field bit
- ✓ 3: Sends a frame with reserved field bit

4.2. Frame Size

- 1: Sends a DATA frame with 2¹⁴ octets in length

... [285 more lines; results/h2spec.txt]

results/testssl.txt

Warning: Target is not a server name: results may be completely wrong, at minimum trust may show false results.

Testing all IPv4 addresses (port 4432): 127.0.0.1

Start 2026-06-25 21:23:50 --> 127.0.0.1:4432 (127.0.0.1) <<--

rDNS (127.0.0.1): localhost.
DNS HTTPS RR (expt.): --
Service detected: HTTP

Testing protocols via sockets except NPN+ALPN

SSLv2 not offered (OK)
SSLv3 not offered (OK)
TLS 1 not offered
TLS 1.1 not offered
TLS 1.2 offered (OK)
TLS 1.3 offered (OK): final
QUIC Local problem: No OpenSSL QUIC support
NPN/SPDY not offered
ALPN/HTTP2 h2, http/1.1 (offered)

Testing cipher categories

NULL ciphers (no encryption)	not offered (OK)
Anonymous NULL Ciphers (no authentication)	not offered (OK)
Export ciphers (w/o ADH+NULL)	not offered (OK)
LOW: 64 Bit + DES, RC[2,4], MD5 (w/o export)	not offered (OK)
Triple DES Ciphers / IDEA	not offered
Obsoleted CBC ciphers (AES, ARIA etc.)	not offered
Strong encryption (AEAD ciphers) with no FS	not offered
Forward Secrecy strong encryption (AEAD ciphers)	offered (OK)

Testing server's cipher preferences

Hexcode	Cipher Suite Name (OpenSSL)	KeyExch.	Encryption	Bits	Cipher Suite Name (IANA/RFC)
---------	-----------------------------	----------	------------	------	------------------------------

SSLv2					
-					
SSLv3					
-					
TLSv1					
-					
TLSv1.1					
-					
TLSv1.2 (server order)					
xc030	ECDHE-RSA-AES256-GCM-SHA384	ECDH 384	AESGCM	256	
	TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384				
xc02f	ECDHE-RSA-AES128-GCM-SHA256	ECDH 384	AESGCM	128	
	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256				
xcca8	ECDHE-RSA-CHACHA20-POLY1305	ECDH 384	ChaCha20	256	
	TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256				
TLSv1.3 (server order)					
x1302	TLS_AES_256_GCM_SHA384	ECDH 253	AESGCM	256	
	TLS_AES_256_GCM_SHA384				
x1301	TLS_AES_128_GCM_SHA256	ECDH 253	AESGCM	128	
	TLS_AES_128_GCM_SHA256				
x1303	TLS_CHACHA20_POLY1305_SHA256	ECDH 253	ChaCha20	256	
	TLS_CHACHA20_POLY1305_SHA256				

Has server cipher order? yes (OK) -- TLS 1.3 and below

Testing robust forward secrecy (FS) -- omitting Null Authentication/Encryption, 3DES, RC4

FS is offered (OK)	TLS_AES_256_GCM_SHA384
	TLS_CHACHA20_POLY1305_SHA256
	ECDHE-RSA-AES256-GCM-SHA384
	ECDHE-RSA-CHACHA20-POLY1305
	TLS_AES_128_GCM_SHA256
	ECDHE-RSA-AES128-GCM-SHA256
KEMs offered	X25519MLKEM768
Elliptic curves offered:	prime256v1 secp384r1 X25519
TLS 1.2 sig_algs offered:	RSA-PSS-RSAE+SHA512 RSA-PSS-RSAE+SHA384
	RSA-PSS-RSAE+SHA256 RSA+SHA512 RSA+SHA384
	RSA+SHA256
TLS 1.3 sig_algs offered:	RSA-PSS-RSAE+SHA512 RSA-PSS-RSAE+SHA384
	RSA-PSS-RSAE+SHA256

Testing server defaults (Server Hello)

```
TLS extensions          "application layer protocol negotiation/#16" "extended master secret/#23" "session ticket/#35"
"supported versions/#43" "key share/#51" "renegotiation info/#65281"
Session Ticket RFC 5077 hint 43200 seconds, session tickets keys seems to be rotated < daily
SSL Session ID support   yes
Session Resumption       tickets: yes, ID: yes
TLS 1.3 early data support no early data offered
TLS clock skew           Random values, no fingerprinting possible
Certificate Compression   none
Client Authentication     none
Signature Algorithm       SHA256 with RSA
Server key size           RSA 2048 bits (exponent is 65537)
Server key usage          --
Server extended key usage --
Serial                   040078C3E8674C6FA48D6F67D0F45DCC1F2C423C (OK: length 20)
Fingerprints              SHA1 FB4AE28D3800D3CC3E3E98244006B9A217B6E0DB
                          SHA256 02E004FD2E25BD18CFD316460E0D5185A05BDD49288FA5FD5A567EF1099CCB68

Common Name (CN)         bench.local
subjectAltName (SAN)     bench.local localhost 127.0.0.1
Trust (hostname)         Ok via SAN
Chain of trust           NOT ok (self signed)
EV cert (experimental)   no
Certificate Validity (UTC) 364 >= 60 days (2026-06-25 19:35 --> 2027-06-25 19:35)
ETS/"eTLS", visibility info not present
Certificate Revocation List --
OCSP URI                 --
                          NOT ok -- neither CRL nor OCSP URI provided
OCSP stapling            not offered
OCSP must staple extension --
DNS CAA RR (experimental) not checked: IP address scan, no domain to query
Certificate Transparency  --
Certificates provided     1
Issuer                   bench.local
Intermediate Bad OCSP (exp.) Ok
```

Testing HTTP header response @ "/"

```
HTTP Status Code         200 OK
HTTP clock skew          0 sec from localtime
Strict Transport Security 730 days=63072000 s, includeSubDomains, preload
Public Key Pinning        --
Server banner             (no "Server" line in header, interesting!)
Application banner        --
Cookie(s)                 (none issued at "/")
Security headers           X-Frame-Options: DENY
                          X-Content-Type-Options: nosniff
                          Permissions-Policy: camera=(), microphone=(),
                          geolocation=(), payment=()
                          Referrer-Policy: strict-origin-when-cross-origin
Reverse Proxy banner      --
```

Testing vulnerabilities

```
Heartbleed (CVE-2014-0160) not vulnerable (OK), no heartbeat extension
CCS (CVE-2014-0224)        not vulnerable (OK)
Ticketbleed (CVE-2016-9244), experimental not vulnerable (OK)
Opossum (CVE-2025-49812)   not vulnerable (OK)
ROBOT                      Server does not support any cipher suites that use RSA key transport
Secure Renegotiation (RFC 5746) supported (OK)
Secure Client-Initiated Renegotiation not vulnerable (OK) -- mitigated (1 successful renege within 10 in 33s(timeout))
CRIME, TLS (CVE-2012-4929) not vulnerable (OK)
BREACH (CVE-2013-3587)     no gzip/deflate/compress/br HTTP compression (OK) - only supplied "/" tested
POODLE, SSL (CVE-2014-3566) not vulnerable (OK), no SSLv3 support
TLS_FALLBACK_SCSV (RFC 7507) No fallback possible (OK), no protocol below TLS 1.2 offered
... [68 more lines; results/testssl.txt]
```

results/proto.env

```
zion_alpn=2
nginx_alpn=2
```

results/zion.log

ZION EDGE GATEWAY – initializing...

ZION edge gateway v0.4.2

```
--- hardware ---
  os          linux / x86_64
  [!!!] cpu    3 cores
  [!!!] ram     2.0 GB
--- crypto / simd ---
[ok] aes-ni / armv8 ce  yes    hardware TLS encrypt
[!!!] sha-256 hw       no     hardware HMAC + cert verify
[!!!] neon            no     ARM SIMD
[ok] avx2             yes    x86 SIMD
--- os networking ---
[ok] so_reuseport      yes    kernel load balance
[ok] tcp_fastopen      yes    0-RTT TCP
[ok] tcp_quickack      yes    linux only – sub-ms ACK
--- cpu cache ---
  line / l1d / l2      64B / 32 KB / 256 KB
  l1 hot entries       81
--- auto-tuning ---
  workers              3
  conn limit           2048
  backlog              8192
  tcp buf              64 KB
```

```
★ TIER B                      score 40/125 ||
decent – adequate for most production loads
AES-128-GCM 2.9M seal/s/core (measured 80ms)
proxy est.  ~42K cached · ~10K dynamic
```

crypto 2/3 · os net 3/3 · cache 32KB · workers 3 · conns 2K
hint only 2048 MB RAM – conn limit capped at 2048. add memory for higher concurrency.
probed in 3267µs · calibrated in 80ms · ZION_BOOT_PLAIN=1 / ZION_BOOT_FAST=1 to override

loaded from /opt/zion-src/benchmarks/baseline/zion-lab.toml

```
tls loaded (min=1.2, alpn=["h2", "http/1.1"])
routes (3)
  /api/{*rest}          → backend
  /_next/static/{*rest} → frontend  static
  /*{*rest}             → frontend
```

xff_mode: Append

watching /opt/zion-src/benchmarks/baseline/zion-lab.toml

monitoring 1 upstreams

listening HTTP on 0.0.0.0:8082

warning: upstream http://127.0.0.1:9090 is DOWN – 503 until it recovers; adaptive re-probe in ~220ms

pre-warmed 1 upstream connections

listening HTTPS on 0.0.0.0:4432

► READY 0.0.0.0:4432 (https) · 0.0.0.0:8082 (http)

live dashboard: zion top --url http://127.0.0.1:8082/_zion/snapshot.json

health: http://127.0.0.1:8082/healthz

upstream http://127.0.0.1:9090 is UP (554us)

[2m2026-06-25T21:23:46.490158Z][0m [33m WARN[0m [2mh2::proto::ping_pong][0m[2m:][0m recv PING ack that we never sent:

Ping { ack: true, payload: [105, 110, 118, 97, 108, 105, 100, 0] }

proxy error: client error (SendRequest)

tls handshake failed from 127.0.0.1:58984: tls handshake eof

tls handshake failed from 127.0.0.1:59018: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:59032: peer is incompatible: SignatureAlgorithmsExtensionRequired

tls handshake failed from 127.0.0.1:59050: peer is incompatible: Tls12NotOfferedOrEnabled

tls handshake failed from 127.0.0.1:59082: peer doesn't support any known protocol

tls handshake failed from 127.0.0.1:59134: peer doesn't support any known protocol

tls handshake failed from 127.0.0.1:41896: peer doesn't support any known protocol

tls handshake failed from 127.0.0.1:41948: peer is incompatible: NoCipherSuitesInCommon

tls handshake failed from 127.0.0.1:41952: peer is incompatible: NoCipherSuitesInCommon

tls handshake failed from 127.0.0.1:41970: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:41986: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:42016: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:57990: peer is incompatible: NoCipherSuitesInCommon

tls handshake failed from 127.0.0.1:57998: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:58026: peer is incompatible: NoCipherSuitesInCommon

tls handshake failed from 127.0.0.1:58040: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:58068: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:58118: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:58178: received fatal alert: CloseNotify

tls handshake failed from 127.0.0.1:46726: peer is incompatible: NoKxGroupsInCommon

tls handshake failed from 127.0.0.1:46740: received fatal alert: CloseNotify

```
tls handshake failed from 127.0.0.1:46762: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:46788: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:46834: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:46864: peer is incompatible: SignatureAlgorithmsExtensionRequired
tls handshake failed from 127.0.0.1:46892: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:46920: peer is incompatible: NoSignatureSchemesInCommon
tls handshake failed from 127.0.0.1:35082: peer is incompatible: NoCipherSuitesInCommon
tls handshake failed from 127.0.0.1:35096: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:35174: peer is incompatible: SignatureAlgorithmsExtensionRequired
tls handshake failed from 127.0.0.1:35200: received corrupt message of type MessageTooShort
tls handshake failed from 127.0.0.1:59020: peer is incompatible: NoKxGroupsInCommon
tls handshake failed from 127.0.0.1:57904: peer is incompatible: NoCipherSuitesInCommon
tls handshake failed from 127.0.0.1:57944: peer is incompatible: NoCipherSuitesInCommon
tls handshake failed from 127.0.0.1:57976: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:57988: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:58018: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:58044: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:58078: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:43438: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:43480: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:43508: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:43550: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:43590: received fatal alert: CloseNotify
tls handshake failed from 127.0.0.1:56384: tls handshake eof
tls handshake failed from 127.0.0.1:41562: tls handshake eof
signal received, draining in-flight connections...
all connections drained cleanly
ZION offline.
```